THE WORLD.

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OUR FOREMOST ELECTRICIAN

The Wonderful Discoveries and Daring Theories of Nikola Tesla, as Told by Him to The World.

GREATER EVEN THAN EDISON.

Hard at Work Experimenting with a New Kind of Light and Promises Wonders That Are Now Undreamed Of by Scientists.

THE ELECTRICITY OF THE FUTURE.

He Predicts It Will Be the Great Labor Saver—Picturesque Personality of a Man Who Is Toying with the Secrets of the Universe.

There is a small country in Europe called Lika, and in that country there is a village of forty houses, called Smiljan.

You have probably never heard of either, but you and many others will live to know about a baby that was born in the village of Smiljan in the country called Lika thirty-seven years ago.

It was probably the usual Slavonic baby, very dark, very nervous and doomed in advance to a dull and wasted life. Lika is really a part of Servia, and Servia had her day long ago. Her men gained fame by fighting and helping the Hungarians to keep Asia out of Europe. Huxley, who thinks seriously, and Oscar Wilde, who is foolish, can enjoy life and London now because those old fighters fought so hard.

Most of Servia's modern babies might as well have avoided the annoyance of birth for all they can ever do in the world, but this baby is an exception.

He was visible yesterday, in his grown-up condition, at Delmonico's. His name is Nikola Tesla. He is the best electrician living, if men who know about electricity may be believed. He is to the average electrician as Horace Greeley is to Bill Nye. He is serious, he is earnest, and in all ways he commands respect.

Every scientist knows his work and every foolish person included in the category of New York society knows his face. He dines at Delmonico's every day. He sits each night at a table near a window. When Ward McAllister, that strange contradiction of the theory that nature abhors a vacuum, wanders in, he sees Nikola Tesla with his head buried in an evening paper.

Every foolish young man who cares for the law of gravitation only because it interferes with jumping fences, every foolish young woman who thinks that there is something new about her two-cent love affair, has seen this serious, owl-faced Servian eating his dinner and thinking about electrical vibrations.

Nikola Tesla is almost the tallest, almost the thinnest and certainly the most serious man who goes to Delmonico's regularly.

VERY THIN, AND VERY TALL.

He has eyes set very far back in his head. They are rather light. I asked him how he could have such light eyes and be a Slav. He told me that his eyes were once much darker, but that using his mind a great deal had made them many shades lighter. I have often heard it said that using the brain makes the eyes lighter in color. Tesla's confirmation of the theory, through his personal experience is important.

He is very thin, is more than six feet tall and weighs less than a hundred and forty pounds. He has very big hands. Many able men do—Lincoln is one instance. His thumbs are remarkably big, even for such big hands. They are extraordinarily big. This is a good sign. The thumb is the intellectual part of the hand. The apes have very small thumbs. Study them and you will notice this.

Nikola Tesla has a head that spreads out at the top like a fan. His head is shaped like a wedge. His chin is as pointed as an ice-pick. His mouth is too small. His chin, though not weak, is not strong enough. His face cannot be studied and judged like the faces of other men, for he is not a worker in practical fields. He lives his life up in the top of his head, where ideas are born, and up there he has plenty of room. His hair is jet black and curly. He stoops—most men do when they have no peacock blood in them. He lives inside of himself. He takes a profound interest in his own work. He has that supply of self-love and self-confidence which usually goes with success. And he differs from most of the men who are written on and talked about in the fact that he has something to tell.

WONDERS OF THE FLY.

It is most pleasing to hear this wise man talk about figures that tire the usual brain. Mr. Delmonico and an aged friend were talking to Mr. Tesla when I saw him night before last. Mr. Tesla having had his supper was discussing, by way of relaxation, the number of vibrations of the wing of a certain fly. The fly, of which Mr. Tesla did not know the name, is a fly that flies straight ahead so fast that you can't see him, and flies back in a straight line without any apparent effort.

"That," said Mr. Tesla, "is the strongest living animal, in proportion to his size. He moves his wings about twenty-five thousand times to the second." The aged man who had consented to take a drink with Mr. Delmonico mopped his forehead worridly, for it was a hot night.

"And that is nothing from the ordinary point of view," mused the distinguished Mr. Tesla. "You see, if you move your hand ten times in a second and then try to move it one hundred times to the second, it will take a hundred times as much strength to move it ten times as fast. The fly that moves its wings twenty-five times to the second requires twenty-five thousand multiplied by twenty-five thousand times as much force as he would to move them once in a second. That makes him use up about six hundred million times as much force as you might think to move his wings like that." Mr. Delmonico's aged friend said: "Don't talk of such exertion in this weather," and wearily mopped himself out of the place.

When the aged man had gone Mr. Tesla said he could go to sleep floating on the water, and Mr. Delmonico, who is a very handsome young man with a pointed black beard, said that he did not believe sharks ever ate any one up. Mr. Tesla said that was nonsense, for in the lower Danube pikes sometimes had seized men and dragged them beneath the surface. The pikes were four feet long, and became so dangerous that swimming about there was prevented. Mr. Delmonico said he was going to Sharon Springs, and started for the Springs at once.

Mr. Nikola Tesla then talked, at my urgent request, about electricity and the things that he hopes to do.

There is no intention here to give a technical account of Mr. Tesla's past achievements and future ambitions. It would be much too hard to write to begin with, and utterly incomprehensible to almost every one after being written. The idea is to discover the new great electrician thoroughly; to interest Americans in the Smiljan baby's personality, so that they may study his future achievements with proper care.

TESLA'S THEORY OF LIGHT.

Mr. Tesla's biggest undertaking at present—that to which he is devoting his most earnest efforts—is the production of light by the vibrations of the atmosphere. He has no intention of heating a bit of cinder red hot and letting it glow by incandescence. The present incandescent system, compared to the Tesla idea, is as primitive as an ox cart with two solid wooden wheels compared to modern railroading.

The light of the sun, according to Mr. Tesla, is the result of vibrations in 94,000,000 miles of ether which separate us from the centre of this solar system. Mr. Tesla's idea is to produce here on earth vibrations similar to those which cause the sunlight, and thus to give us a light as good as that of the sun, with no danger from clouds or other obstructions. Mr. Tesla has already achieved decided success in this line. He takes in his hand a long bar of glass, which, by vibration alone, lights up into most amazing brilliancy. He himself comes out of his experiments a most radiant creature, with light flaming at every pore of his skin, from the tips of his fingers and from the end of every hair on his head.

In explaining his experiments, Mr. Tesla uses figures calculated to pulverize an ordinary mind.

"It is difficult for me," he said, "to give you an idea that you will readily grasp about this question of vibration. In ordinary life our minds do not deal with the figures that come up in such investigations, but take a 5 and put after it fourteen zeroes; then you will have the number of vibrations which occur in the ether every second and which produce light."

I carried out Mr. Tesla's suggestion, with the following result—500,000,000,000,000.

"All I have to do," said Mr. Tesla, "to duplicate the sunlight is to get this number of vibrations to the second with my machinery on earth. I have succeeded up to a certain point, but am still at work on the task."

"THE POWER OF VIBRATION.

I carried in various ways to present in cold writing some notion of what five hundred trillions of vibrations to the second might mean. I didn't succeed very well. The nearest I could was to mention, at Mr. Tesla's suggestion, the following fact: If a mass of metal as big as the Delmonico restaurant, in which we sat, possessing 10,000 times the resisting force of the most finely tempered steel, should be caused to vibrate with one millionth of the rapidity of the light-producing electric vibrations in ether, that mass of metal, 10,000 times harder than steel, would simply vanish into the air like smoke. It would disappear into separate atoms too small to be seen and would never be heard of again.

Electricity in its vibrations, according to Mr. Tesla, has a great advantage over all other things, such as flies' wings and other material bodies. Electricity has no weight, and therefore no opposition is offered to its moving backward and forward freely any number of times to the second.

"It is perfectly easy to prove that electricity weighs nothing," said Mr. Tesla. "I will load you so full of electricity that you can't hold any more and then put you on the finest weighing machine, and you will not find one-thousandth part of an ounce added to your weight."

I ventured to suggest to Mr. Tesla that as a vibrator electricity might meet with serious competition from other statesmen, but his mind was so serious that he only said: "No statesman could vibrate fast enough to be of any value scientifically."

Before deciding to give Mr. Tesla unlimited space in the greatest newspaper that was ever heard of, I had heard from a dozen reliable sources that there was not the slightest doubt about his being a very great man. Authorities unite in declaring him the very greatest man living in the line of abstract electrical research.

One able electrician told me that there never was such a man for working out purely intellectual problems. Another said he could conceive of nothing more extraordinary than the devotion and admiration for Tesla entertained by all of the young electrical engineers. It is a pitiful thing for a morbidly conscientious writer to find himself booming one not altogether worthy of his periods and climaxes, but I am certain in the case of Mr. Tesla that it is safe to go ahead.

ONE OF HIS DISCOVERIES.

Mr. Tesla discovered the rotating magnetic field. That seems to me, next to his idea of getting light by vibration, the best thing he ever did. The rotating magnetic field is a thing which may be described but not understood. Everybody knows that a magnet will seize a piece of iron and hold it firmly; everybody knows that the magnet must use up force in holding that iron, but of course as long as it holds the iron perfectly still the force is wasted. The piece of iron if left alone would stand still. There is no use in getting a magnet to make it stand still. But Mr. Tesla found that he could get a magnet to use its force in such a way as to cause the piece of iron to spin violently round and round. He can make a wheel at a distance from the source of electro-magnetic force spin round with 10,000 horse-power. He expects to apply this principle in employing the strength of the Niagara Falls electric current. The fact may be mentioned that the Niagara Falls people who have relied upon Tesla to tell them how they may use their power at a distance have adopted his scheme.

NIKOLA TESLA.

Showing the Inventor in the Effulgent Glory of Myriad Tongues of Electric Flame After He Has Saturated Himself with Electricity.

This rotating magnetic field struck me as a most impressive discovery. I asked Mr. Tesla whether he didn't think it possible that the spinning, rotary motion of the earth and her fellow planets, commonly attributed to some unexplained primary propulsive force, might be due to the application on a grand scale of his rotating magnetic field idea. I suggested that the sun might be a great magnet, that the five hundred trillion vibrations per second which he spoke of showed considerable electrical power somewhere; that there was a great deal of iron and a powerful lot of electricity in the earth. Mr. Tesla observed that he thought it was dangerous to jump at such conclusions, but he treated my enthusiasm, born of complete ignorance, with a kind toleration, which did much to convince me of his true greatness.

THE SECRET OF ELECTRICITY.

Electricians in general think that Tesla's best work thus far is a machine which has industrial value as a new and more direct agent for producing electrical force. I didn't care much about that compared with the rotating magnetic field, but I asked Mr. Tesla to tell me in as few words as possible just how we get electricity, and why we get it when we do get it. Said Mr. Tesla:

"We get electricity by causing a wire to revolve near a magnet. The stronger the magnet the faster the revolutions of the wire, and the bigger the wire the more electricity.

"Why we get electricity in this way, and what electricity is, are different questions. Every electrician has his theory. I have one which I think I can demonstrate mathematically. There is no accepted explanation of the most extraordinary phenomenon in nature."

Mr. Tesla does not care to see in cold print an account of those things which he hopes to accomplish, or to see accomplished, by means of electricity.

"You would think me a dreamer and very far gone," he said, "if I should tell you what I really hope for. But I can tell you that I look forward with absolute confidence to sending messages through the earth without any wires. I have also great hopes of transmitting electric force in the same way without waste. Concerning the transmission of messages through the earth I have no hesitation in predicting success. I must first ascertain exactly how many vibrations to the second are caused by disturbing the mass of electricity which the earth contains. My machine for transmitting must vibrate as often to put itself in accord with the electricity in the earth."

TOOK 250,000 VOLTS HIMSELF.

Mr. Tesla is the interesting person who, in Philadelphia, before a large gathering, allowed a quarter of a million volts of electricity to go through his body. Having seen Carlyle Harris and one other unfortunate individual instantly killed by the application of less than 2,000 volts, I asked Mr. Tesla if he didn't feel a little worried about taking a current of a quarter of a million volts. Said he:

"I did at first feel apprehensive. I had reasoned the thing out absolutely, nevertheless there is always a certain doubt about the practical demonstration of a perfectly satisfactory theory. My idea of letting this current go through me was to demonstrate conclusively the folly of popular impressions concerning the alternating current. The experiment had no value for scientific men. A great deal of nonsense is talked and believed about 'volts,' &c. A million volts would not kill you or hurt you if the current vibrated quickly enough—say half a million times to the second. Under such conditions the nerves wouldn't respond quickly enough to feel pain."

"You see, voltage has nothing to do with the size and power of the current. It is simply the calculation of the force applied at a given point. It corresponds to the actual pressure per square inch at the end of a water pipe, whether the volume of the water be great or small. A million volts going through you doesn't mean much under proper conditions. Imagine a needle so small that the hole it would make in going through your body would not allow the blood to escape. Imagine it so small that you couldn't even feel it. If you had it put through your arm slowly, that would be, electrically speaking, a very small voltage. If you had it stuck through your arm with great rapidity, going, say, at the rate of a hundred miles a second, that would be very high voltage. Voltage is speed, pressure at a given point. It wouldn't do you any more harm to have a needle shot through your arm very rapidly—that is to say, with high voltage—than it would to put it through slowly. In fact, if it hurt you at all, the slow operation would probably hurt more than the other. The question of danger is simply the size of current, and yet if a big enough current should be turned against you and broken with sufficient rapidity—if it should, so to speak, jerk back and forth an inconceivable number of times to the second—it wouldn't kill you. Whereas, if applied continuously, it would simply burn you up."

ELECTRICITY OF THE FUTURE.

When Mr. Tesla talks about the electrical problems upon which he is really working he becomes a most fascinating person. Not a single word that he says can be understood. He divides time up into billionths of seconds, and supplies power enough from nothing apparently to do all the work in the United States. He believes that electricity will solve the labor problem. That is something for Mr. Debs to ponder while he languishes in his dungeon. It is certain, according to Mr. Tesla's theories, that the hard work of the future will be the pressing of electric buttons. A few centuries from now the criminal instead of working a treadmill or picking oakum will be sentenced to press fifteen electric buttons every day. His fellows, long since disused to work, will look upon his toil with pity and horror.

Mr. Nikola Tesla is to be envied. He owns one of those rare minds which do not absorb trivialities. A computation of the vibrations of a fly's wing is to him what croquet would be to Grover Cleveland—it is play. He can play when he wants to, but he doesn't do it very often.

Mr. Delmonico lowers his voice when he speaks of Mr. Tesla, as Boston cabdrivers used to lower their voices in speaking of John L. Sullivan. He said:

"That Tesla can do anything. We managed to make him play pool one night. He had never played, but he had watched us for a little while. He was very indignant when he found that we meant to give him fifteen points. But it didn't matter much, for he beat us all even and got all the money. There are just a few of us who play for 25 cents, so it wasn't the money we cared about, but the way he studied out pool in his head, and then beat us, after we had practised for years, surprised us."

I talked to this Mr. Tesla of Smiljan until all but one of the lights had gone out and until the feeble daylight found Mr. Delmonico's scrub-ladies scrubbing his marble floor. All that he said was interesting, both the electrical things and the others.

HARD WORK HIS PLEASURE.

He is very proud of his Slavonic race. He believes that the poetry of the Slave written in Servian and Croatian would surprise the civilized world if it could be presented in widely known languages.

He says that the thin man relies constantly on the food which he eats at the moment. He looks upon good food as most important and feels the strengthening effects of it within twenty minutes. He believes that work never hurts, but that play does.

Nobody, he thinks, can work enough to hurt him.

He thinks that marriage and love interfere with success.

That suppers are very bad for one in New York, but all right in Paris.

He does not believe in telepathy, which is, according to its exponents, a sort of psychical electricity enabling one mind to communicate ideas to another without words. He considers that what is usually taken as an evidence of the existence of telepathy is mere coincidence. But the working of the human mind through observation and reason interests and amazes him, as it well may.

"Suppose I made up my mind to murder you," he said, "in a second you would know it. Now, isn't that wonderful? By what process does the mind get at all this?"

One wise man whom I knew used to say that the scientists with their jumble of laws governing the universe were ignoramuses, and that there weren't more than half a dozen fundamental laws all told. I asked Mr. Tesla what he thought about that.

"I think," said he, "that they could all be reduced to one."

ARTHUR BRISBANE.